

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

(Not to be reproduced without permission of BIS or used as an Indian Standard)

भारतीय मानक मसौदा
डेंटल स्केलर के विनिर्देश, क्रमांक SC 1/2, 3/4 और 5/6
(IS 4714 का दूसरा पुनरीक्षण)

Draft Indian Standard

Specification for Scalars, Dental, No. SC 1/2, 3/4 and 5/6
(Second Revision of IS 4714)

[ICS 11.060.20]

Dentistry Sectional Committee, MHD 08

Last date for comments:
11 Sep 2026

FOREWORD

(Formal clauses will be added later)

This Indian Standard was originally published in 1968 and the first revision was published in 1985. The second revision of this standard aligns the cross references to the latest standards and revised certification clauses.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test or analysis, shall be rounded off in accordance with IS 2: 2022 'Rules for rounding off numerical values (second revision)'.

1 SCOPE

This standard specifies materials, dimensions and other requirements of scalers No. SC 1/2, 3/4 and 5/6 used in dentistry.

2 REFERENCES

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

<i>IS No</i>	<i>Title</i>
IS 3887 : 2024	Cutting Type Dental Instruments — General Requirements (<i>Second Revision</i>)
IS 2428: 2024 ISO 513 : 2012	Classification and Application of Hard Cutting Materials for Metal Removal with Defined Cutting Edges — Designation of the Main Groups and Groups of Application (<i>Third Revision</i>)
IS 1501 (Part 1) : 2020 ISO 6507-1 : 2018	Metallic Materials — Vickers Hardness Test Part 1 Test Method (<i>Fifth Revision</i>)
IS 7531 : 1990	Surgical Instruments - Corrosion Resistance Of Stainless Steel Surgical Instruments - Methods Of Tests (<i>First Revision</i>)
IS 4905 : 2015 ISO 24153 : 2009	Random Sampling and Randomization Procedures (<i>First Revision</i>)

3 MATERIAL

3.1 The material shall conform to Clause 5 of IS 3887.

3.2 Carbide Tip

The Carbide tip of the scalers shall conform to designation M10 of IS 2428.

4 SHAPE AND DIMENSIONS

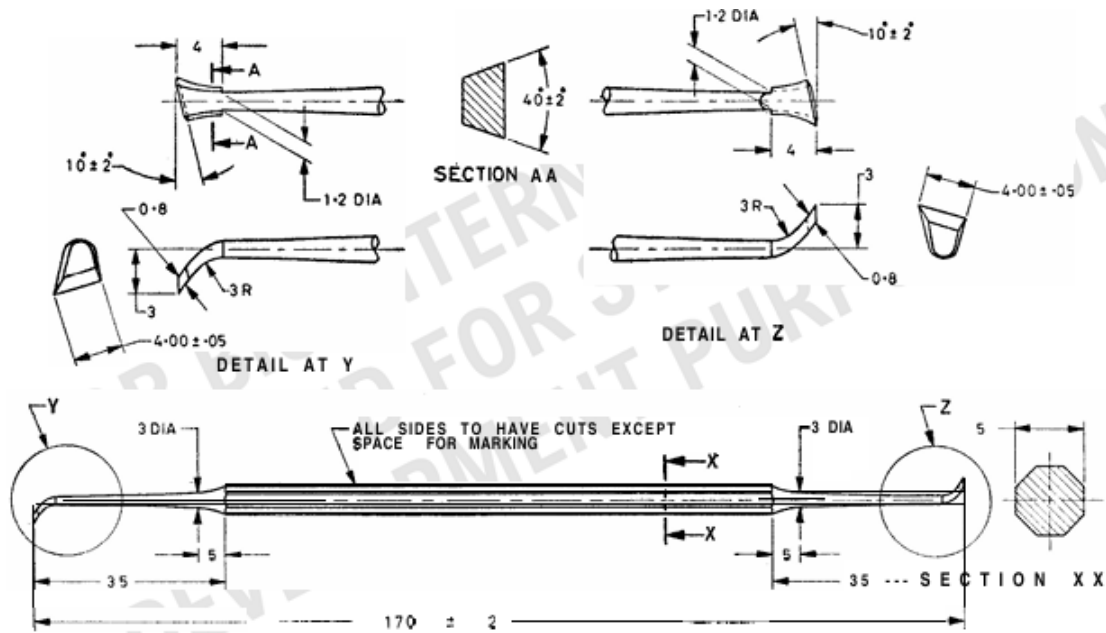
The shape and dimensions shall be as shown in Fig. 1 to 3.



Fig.1, Scaler, Dental , No. SC 1/2



Fig.2 Scaler, Dental, No. SC 3/4



All dimensions in millimetres
Fig.3 Scaler, Dental , No. SC 5/6

5 REQUIREMENTS

The requirements specified in Clause 4 of IS 3887 shall apply.

6 WORKMANSHIP AND FINISH

The requirements specified in Clause 7 of IS 3887 shall apply.

7 HEAT TREATMENT

The scalers except the carbide tip shall be uniformly hardened and tempered to a hardness of 430 to 490 HV , when tested in accordance with IS 1501(Part 1).

8 TESTS

8.1 Practical Test

The working end of the instrument shall be applied to the top of a compression type spring balance at an angle of 20° approximately. A force of 3 kgf (30N approximately) shall be applied for 30 seconds while gripping the handle at about the middle of its length. On releasing the force ,the instrument shank shall not show any sign of damage or acquire a permanent set.

8.2 Elasticity Test

The handle of the instrument shall be gripped in a vice so that the whole length of the shank protrudes out of vice jaw. A force of 3 kgf (30 N approximately) shall be applied at a distance of 2 mm from the end in a direction at right angles to the plane of the working end. On releasing the force, the instrument shall not show any sign of damage or acquire a permanent set. The test shall be repeated on the other end also.

8.3 Corrosion Resistance Test

The scalers shall be tested in accordance with the test specified in IS 7531. It shall show no sign of corrosion after the test.

8.4 Performance Test

The working end of the scaler shall be made to scrape a piece of ivory mounted on a sealing wax platform with moderate force of hand, ten times. The scaler shall do this neatly and shall not show any sign of damage after the test.

9 SAMPLING

Sampling procedure and acceptance criteria for dental scalers shall be as agreed to between the purchaser and the supplier. A recommended scheme for the same is given in Appendix A.

10 MARKING

The space on the handle shall be marked with the number of instrument; manufacturer's name, initials or registered trade-mark; the letters 'SS', and the country of manufacture.

11 PACKING

11.1 Each scaler shall be wrapped either in a polyethylene bag or a wax paper. It shall then be packed in cartons in accordance with the current trade practice. The carton shall bear the name of the scaler; manufacturer's name, initials or registered trade-mark; the words 'Stainless Steel' and the country of manufacture.

11.2 Alternatively, the instruments may be packed as agreed to between the purchaser and the supplier.

11 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.

APPENDIX A
(Clause 9)

**SAMPLING SCHEME AND CRITERIA FOR CONFORMITY
FOR SCALERS, DENTAL**

A-1. Lot

In any consignment, all the scalers of the same materials under similar conditions and having the same surface finish shall constitute a lot.

A-2. Scale of Sampling

The number of scalers to be selected from each lot shall depend upon the size of the lot and shall be in accordance with col 1 and 2 of Table 1.

TABLE 1 SCALE OF SAMPLING
(Clauses A-2 and A-3)

Lot Size (1)	Sample Size (2)
Up to 15	2
16 to 50	3
51 to 150	5
151 and above	8

A-2.1 These scalers shall be selected from the lot at random and in order to ensure randomness of selection, procedures given in IS 4905.

A-3. Number of Tests and Criteria for Conformity

All the scalers selected at random in accordance with col 1 and 2 of Table 1 shall be tested for shape and dimensions, workmanship and finish, heat treatment, practical test, elasticity test, corrosion resistance test, and performance test. The lot shall be considered as conforming to these requirements if none of the scalers in the sample is found to be defective in any of these tests.